

Seminar Abstract

**THE RISE OF AI-POWERED CYBER
CRIME: DEEPPAKES, VOICE CLONING,
AND AUTOMATED PHISHING ATTACKS**

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Abstract

The rapid advancement of artificial intelligence has significantly transformed the cybercrime landscape, enabling attackers to execute highly sophisticated and convincing attacks. This seminar examines the rise of AI-powered cybercrimes, focusing on three major threat categories: deepfake technology, voice cloning attacks, and automated phishing. These technologies exploit human trust and expose critical weaknesses in traditional authentication and verification mechanisms.

The seminar discusses the technical foundations of deepfakes, voice cloning, and AI-generated phishing, highlighting how these technologies are used to impersonate identities, manipulate communication, and conduct large-scale social engineering attacks. It further analyses the growing impact of such attacks on individuals, organisations, and digital trust.

In addition, the seminar presents detection challenges and defence strategies, including technical safeguards, organisational security practices, and individual awareness measures. Legal and regulatory challenges associated with AI-powered cybercrimes are also briefly addressed. By examining emerging threats and mitigation approaches, the seminar emphasises the urgent need for adaptive and multi-layered security solutions in an era where traditional verification methods are increasingly unreliable.

Seminar Outline

1. Introduction

- Overview of cybercrime in the AI era
- How Artificial Intelligence is changing attack strategies
- Three main threats: Deepfakes, Voice Cloning, AI Phishing

2. Deepfake Technology

2.1 How Deepfakes Work

- Introduction to GANs (Generative Adversarial Networks)
- Face manipulation and facial landmark detection

2.2 Real-World Case Study: Deepfake Financial Fraud

- Impersonation using AI-generated video
- Use of fake video conferencing for social engineering
- Financial loss and trust exploitation

2.3 Detection Challenges

- Human difficulty in detecting deepfakes
- AI-based deepfake detection accuracy limitations
- Continuous evolution of deepfake realism

3. Voice Cloning Attacks

3.1 Voice Cloning Technology

- How it works
- Text-to-Speech AI models
- Mimics accent, tone, emotions

3.2 Real-World Case Study: Voice Impersonation Fraud

- Executive voice cloning for financial scams
- Bypassing traditional voice authentication
- Impact on organisations and individuals

3.3 Current Threat Landscape

- Growth of vishing and voice-based fraud
- Targeting of elderly users and corporate employees
- Limitations of voice-based security systems

4. AI-Powered Phishing

4.1 AI Enhancement of Phishing

- Traditional phishing vs AI-generated phishing
- Use of large language models for email generation
- Increased success rates due to personalisation

4.2 Real-World Case Study: AI-Assisted Social Engineering

- AI-generated phishing leading to ransomware attacks
- Business Email Compromise incidents
- Financial and operational damage

4.3 Scale of Problem

- Global rise in AI-driven phishing attacks
- Availability of phishing tools on the dark web

5. Defence Strategies Against AI-Powered Cyber Crimes

5.1 Technical Detection and Prevention

- Deepfake detection techniques
- Voice verification safeguards
- Email and network security solutions

5.2 Organisational Protection Measures

- Multi-factor authentication
- Verification and callback procedures
- Cyber security awareness training

5.3 Individual Protection Measure

- Verification before acting on urgent requests
- Limiting public sharing of voice and video data
- Importance of scepticism and awareness

6. Legal Framework & Future

6.1 Legal and Regulatory Aspects

- Overview of existing cybercrime laws
- Challenges in regulating AI misuse Gaps in current legal frameworks

6.2 Future Threats

- Real-time deepfake attacks
- AI-as-a-Service cybercrime platforms
- Increasing sophistication of AI attacks

7. Conclusion

- Summary of AI-powered cybercrime threats
- Limitations of traditional security mechanisms
- Importance of adaptive and continuous security
- Need for awareness, verification, and proactive defence

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RELEVANCE OF THE TOPIC

- Rapid growth of AI has enabled sophisticated cyber crimes
- Deepfakes and voice cloning enable realistic impersonation
- Automated phishing increases attack scale and success rate
- Threatens digital trust, privacy, and financial security
- Highly relevant in the era of generative AI and digital communication

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SCOPE AND SIGNIFICANCE

- Covers AI-driven cyber threats affecting individuals and organizations
- Applicable to banking, governance, social media, healthcare, and enterprises
- Highlights emerging risks beyond traditional cyber attacks
- Helps in understanding the need for advanced detection and regulation
- Supports awareness, prevention, and future cybersecurity research

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OUTLINE

- Overview of AI-Powered Cyber Crime
- Evolution of Cyber Attacks Using AI
- Deepfake Technology and Criminal Misuse
- Voice Cloning Based Fraud Attacks
- Automated Phishing Using Generative AI
- Real-World Impact and Case Examples
- Detection, Prevention, and Mitigation Techniques
- Legal and Ethical Challenges
- Future Trends in AI-Driven Cyber Crime

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THANK YOU

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